lecture 17.

PARTIAL ORDER.

- Reflexive
- Anti Symmetric
- Transitive.

$\leq$.

Ex 2 504. $R = \{(a,b) \mid a \geq b\}$ $\subseteq$. $A = \mathbb{Z}$.

1 Reflexive $\forall a \in A$ $(a,a) \in R$, $A = \mathbb{Z}$.

$\forall a \in \mathbb{Z}$ $a \geq a$ ✓.

2- AntiSymmetric $\forall a,b \in A$ if $(a,b) \in R \wedge (b,a) \in R \to a = b$.

$\forall a,b \in \mathbb{Z}$. if $a \geq b \wedge b \geq a \to a = b$.

3- Transitive $\forall a,b,c \in A$ if $(a,b) \in R \wedge (b,c) \in R \to (a,c) \in R$.

$\forall a,b,c \in \mathbb{Z}$. if $a \geq b \wedge b \geq c \to a \geq c$ ✓.

hence Partial Order.

Ex 2 504 $R = \{(a,b) \mid a \equiv b \bmod m\}$ $A = \mathbb{Z}$. $m > 1 \wedge m \in \mathbb{Z}$.

1 Reflexive $\forall a \in A$ $(a,a) \in R$, $A = \mathbb{Z}$.

$\forall a \in \mathbb{Z}$ $a \equiv a \bmod m$. ✓.

2- AntiSymmetric $\forall a,b \in A$ if $(a,b) \in R \wedge (b,a) \in R \to a = b$.

$\forall a,b \in \mathbb{Z}$ if $a \equiv b \bmod m \wedge b \equiv a \bmod m$

X.

3- Transitive $\forall a,b,c \in A$ if $(a,b) \in R \wedge (b,c) \in R \to (a,c) \in R$.

Ex 2 $R = \{(a,b) \mid a \subseteq b\}$ $P(A)$

1 Reflexive $\forall a \in A \quad (a,a) \in R$.

$\forall a \in P(A) \quad a \subseteq a$ ✓.

2- AntiSymmetric $\forall a,b \in A$ if $(a,b) \in R \wedge (b,a) \in R \rightarrow a=b$.

$\forall a,b \in P(A)$ if $a \subseteq b \wedge b \subseteq a \rightarrow a=b$.

3- Transitive $\forall a,b,c \in A$ if $(a,b) \in R \wedge (b,c) \in R \rightarrow (a,c) \in R$.

$\forall a,b,c \in P(A)$ if $a \subseteq b \wedge b \subseteq c \rightarrow a \subseteq c$

$\underbrace{\text{Po}}\underbrace{\text{SET}}$.

Partial order.

$(S, \preceq)$

$(P(A), \subseteq)$. $\longleftrightarrow$ $R=\{(a,b) \mid a \subseteq b\}$ $P(A)$

$(a,b) \in R$.

$a \preceq b$.

$(Z, \geq)$. $\longleftrightarrow$ $R=\{(a,b) \mid a \geq b\}$. $A=Z$.

$(Z, \leq)$

$(Z, =)$

Comparable: Two element "a, b" $\in S$ in $(S, \preceq)$ are Comparable if: $a \preceq b \vee b \preceq a$.

Ex5 / 584. if 5, 7 are Comparable in $(Z^+, /)$.

$5 \preceq 7 \vee 7 \preceq 5$.

$(5,7) \in R \vee (7,5) \in R$.

$5/7 \vee 7/5$

$F \vee F = F$.

Total Order:- if $\forall a,b \in S \quad a \preceq b \vee b \preceq a$.

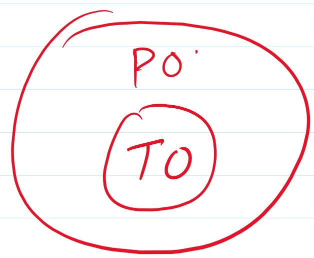
Ex 7:- $(Z^+, /)$. Not a TO
SoS.

why $3 \preceq 7 \vee 7 \preceq 3$.
$(3,7) \in R \vee (7,3) \in R$.

$3/7 \vee 7/3$.
$F \vee F = F$.

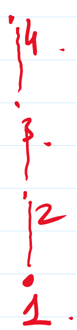
Ex 6:- $(Z, \le)$
SoS

$-\infty \le -\infty+1 \le -\infty+2 \le \dots \quad -2 \le -1 \le 0 \le 1 \le \dots \le +\infty$.



Hasse Diagram.

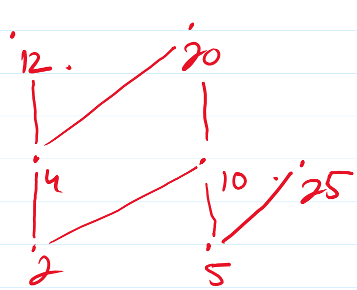
$(\{\cancel{1},2,3,\cancel{4}\}, \le)$.



Total Order.

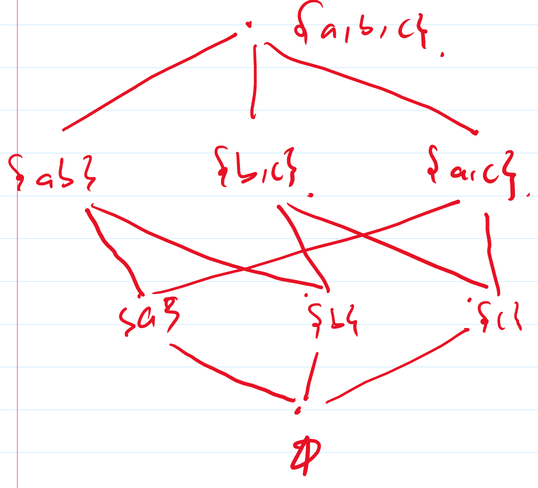
$R = \{\cancel{(1,1)}, (1,2), (1,3), (1,4)$
$\cancel{(2,2)}, (2,3), (2,4),$
$\cancel{(3,3)}, (3,4)$
$\cancel{(4,4)}\}$.

Ex 14 $(\{\cancel{2}, 4, \cancel{5}, 10, \cancel{12}, \cancel{20}, \cancel{25}\}, /)$.



$R = \{\cancel{(2,2)}, (2,4), (2,10), (2,12), (2,20),$
$\cancel{(4,4)}, (4,12), (4,20),$
$\cancel{(5,5)}, (5,10), (5,20), (5,25),$
$\cancel{(10,10)}, (10,20),$
$\cancel{(12,12)},$
$\cancel{(20,20)},$
$\cancel{(25,25)}\}$.

$(P\{a,b,c\}, \subseteq)$.



$P(\{a,b,c\}) = \{\cancel{\phi}, \{a\}, \{b\}, \{c\}, \{a,b\}, \{b,c\}, \{a,c\}, \{a,b,c\}\}$.

$R = \{(\phi,\phi), (\phi,\{a\}), (\phi,\{b\}) \ldots$

$\cancel{(\{a\},\{a\})}, (\{a\},\{a,b\}), (\{a\},\{a,c\}), \ldots$

$\cancel{(\{b\},\{b\})} \ldots$

$\cancel{(\{c\},\{c\})} \ldots$

$\cancel{(\{a,b\},\{a,b\})}, (\{a,b\},\{a,b,c\}),$

$\vdots$

$\cancel{(\{a,b,c\},\{a,b,c\})}\}$.

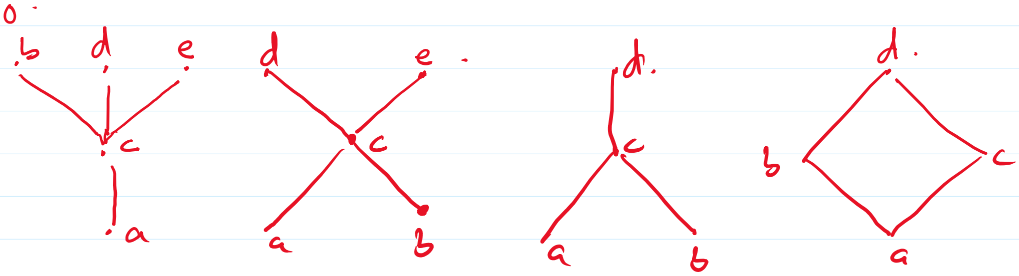
1) Maximal 'a' $\in S$ is Maximal in $(S, \preceq)$.
iff $\neg\exists_b \; a \prec b$.

2) Minimal 'a' $\in S$ is Minimal in $(S, \preceq)$
iff $\neg\exists_b \; b \prec a$.

3) Greatest $a \in S$ is Greatest in $(S, \preceq)$
iff $\forall_{b \in S} \; b \preceq a$

4). Least. $a \in S$ is least in $(S, \preceq)$.
iff $\forall_{b \in S} \; a \preceq b$.

Ex 17.
510.



	(1)	(2)	(3)	(4)
Maximal	b, d, e.	d, e	d.	d.
Minimal	a	a, b.	a, b.	a
Greatest	—	—	d.	d
Least.	a	—	—.	a.

Lower Bound

Greatest LB.

Upper Bound

Least Upper Bound.

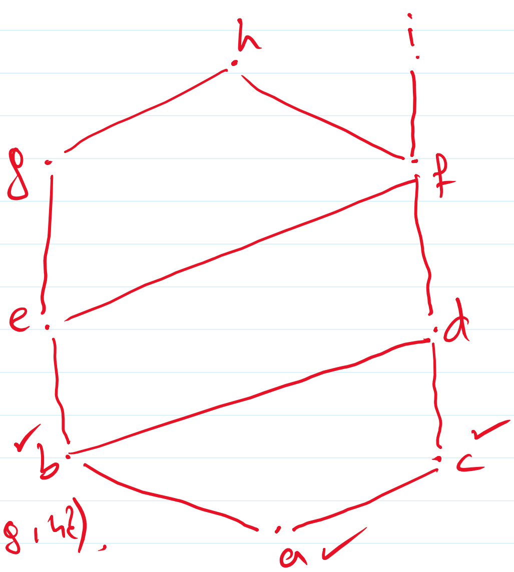
Ex 18
511

UB({g, f}) = {h}.

LUB = h.

LB({g, h}) = {e, b, a}

GreatestLB({g, h}) = e.



UB({a, b, c}) = {d, f, h, i}.

Least UB ({a, b, c}) = d.

LB ({a, b, c}) = {a}.

Greatest LB = a